

LAC SAINT-PIERRE, QC, Canada

WMO: 711980

Lat: 46.183N

Lon: 72.917W

Elev: 53

StdP: 14.67

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-11.2	-6.2	-21.4	1.7	-8.5	-15.9	2.3	-4.0	36.7	27.2	31.2	24.4	9.6	250	0.685

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.4	82.2	70.1	79.7	68.9	77.5	67.7	73.0	78.7	71.5	77.0	70.1	75.5	13.5	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
71.1	115.0	76.7	69.5	108.8	75.4	68.0	103.1	74.0	36.7	78.9	35.4	77.2	34.1	75.7	79.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
30.0	26.8	24.0	-15.1	86.7	5.7	3.4	-19.2	89.1	-22.6	91.1	-25.8	93.1	-29.9	95.6
			-17.5	75.1	4.5	1.7	-20.7	76.3	-23.3	77.3	-25.8	78.3	-29.0	79.6
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	43.7	14.6	16.2	26.7	40.7	54.2	64.5	69.3	67.6	61.2	48.5	36.2	23.3
	DBStd	21.11	12.42	10.98	10.23	8.00	8.01	6.17	5.20	5.63	6.73	7.34	8.63	10.73
	HDD50	4468	1096	945	722	293	46	0	0	0	4	118	416	828
	HDD65	8145	1561	1365	1187	730	343	82	20	38	151	512	863	1293
	CDD50	2186	0	0	1	13	178	436	598	547	339	72	2	0
	CDD65	386	0	0	0	0	9	68	152	120	36	1	0	0
	CDH74	1507	0	0	0	1	48	281	659	408	110	0	0	0
	CDH80	193	0	0	0	0	6	41	100	37	9	0	0	0
	WSAvg	12.0	11.8	11.4	11.9	12.4	11.9	11.3	10.8	10.9	11.7	13.2	13.3	13.0
	PrecAvg	39.6	2.9	2.2	2.4	3.3	3.3	3.7	4.3	3.7	3.4	3.9	3.3	3.2
(15) Precipitation	PrecMax	48.3	4.6	3.9	4.2	5.7	6.5	7.2	7.5	7.6	7.4	6.4	5.6	5.6
	PrecMin	30.2	1.3	1.2	1.0	0.6	1.5	1.7	2.6	1.5	2.2	1.1	1.0	0.9
	PrecStd	4.8	1.0	0.7	0.8	1.4	1.3	1.5	1.2	1.3	1.2	1.5	1.0	1.4
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% DB	41.9	40.5	52.6	65.9	79.2	84.3	86.6	83.6	81.0	71.2	59.4	46.4	
	0.4% MCWB	40.2	35.9	44.7	52.6	63.8	70.6	72.1	72.4	72.0	63.2	54.1	44.1	
	2% DB	38.2	37.2	44.5	60.4	73.9	80.1	82.7	80.5	77.2	66.2	53.6	42.0	
	2% MCWB	36.1	34.4	39.0	49.4	60.1	67.6	70.2	69.7	68.7	59.0	49.1	40.0	
	5% DB	35.6	35.0	41.4	54.9	69.8	77.2	80.0	78.2	74.1	62.1	49.9	39.1	
	5% MCWB	33.9	33.0	36.9	46.4	57.6	65.4	69.6	68.4	67.1	56.3	46.0	37.0	
	10% DB	32.1	32.6	38.8	51.1	65.8	74.2	77.5	76.0	71.4	59.1	46.7	36.6	
	10% MCWB	30.7	31.0	35.0	43.3	56.1	64.0	68.5	67.5	64.7	54.2	43.3	34.5	
	0.4% WB	40.9	38.0	48.7	57.5	68.4	72.7	75.7	75.0	73.3	64.4	55.3	44.5	
	0.4% MCDB	41.1	39.2	51.4	65.6	76.3	81.0	82.2	80.9	78.2	69.5	58.0	45.6	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2% WB	36.4	35.0	40.7	51.6	64.9	70.5	73.5	72.4	70.8	60.7	50.4	40.4	
	2% MCDB	37.8	36.5	44.5	58.1	71.0	77.3	79.2	77.4	75.3	64.2	52.9	41.9	
	5% WB	34.0	33.5	37.8	48.1	61.2	68.5	71.9	70.8	68.5	57.6	46.9	37.4	
	5% MCDB	35.4	35.0	40.9	54.3	67.3	74.3	77.5	75.7	73.3	61.6	49.8	38.8	
	10% WB	30.8	31.3	35.9	45.1	58.1	66.6	70.4	69.3	66.0	55.2	44.3	34.9	
	10% MCDB	32.2	33.3	38.7	50.4	63.9	71.8	76.0	74.4	70.7	59.0	47.2	36.4	
	MDBR	13.2	14.6	12.8	12.3	13.4	13.2	12.4	12.4	12.7	10.6	9.3	9.8	
	5% DB MCDBR	14.9	13.1	13.1	18.6	18.9	16.6	14.7	14.2	14.9	14.8	13.5	10.0	
	5% DB MCWBR	13.8	12.6	10.0	12.0	10.0	8.4	7.8	8.1	9.8	10.1	11.5	10.1	
	5% WB MCDBR	13.5	13.0	12.6	17.3	16.2	13.9	12.6	12.2	14.3	13.4	12.7	9.6	
	5% WB MCWBR	13.7	12.7	10.3	12.3	11.4	8.8	7.7	8.3	10.1	10.3	11.9	10.1	
(40) Clear-Sky Solar Irradiance	taub	0.257	0.285	0.318	0.355	0.382	0.419	0.421	0.410	0.376	0.335	0.308	0.281	
	taud	2.336	2.242	2.281	2.359	2.354	2.250	2.254	2.288	2.371	2.492	2.478	2.335	
	Ebn at Noon	271	282	288	286	280	269	267	265	265	263	250	245	
	Edh at Noon	23	32	36	36	38	43	42	39	33	25	21	21	
(44) All-Sky Solar Radiation	RadAvg	452	739	1147	1415	1627	1782	1767	1551	1218	710	424	308	
	RadStd	59	82	101	147	166	131	151	72	99	58	54	26	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (43 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page